AL SAFA 2 PARK

Dubai Municipality — AI-Powered Park Design Challenge

DELIVERABLE 5

Key Sections & Elevations

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Schematic analytical diagrams, true 1:1 scale, no vertical exaggeration — not construction-ready.

5.1 Overview

Three cross-sections through the confirmed Preliminary Design Masterplan (Deliverable 4), chosen to show the site's most vertically distinct conditions: the arrival-to-buffer sequence, the mosque-buffer-to-active-court graduation, and the shade-canopy rhythm along the promenade.

5.2 Ground Line: Confirmed Flat, Not Assumed

[VERIFIED] Every entity in the source CAD sits at Z=0 (Deliverable 1). Rather than take that as sufficient on its own, an independent elevation-API check was run at 10 points across the site. Result: a clean step pattern — 7 points at exactly -2.0 m, 3 points (clustered near the NE corner) at exactly 3.0 m. Two suspiciously round values splitting cleanly along a line through a 90 m-wide site is the signature of a coarse global DEM's pixel-grid boundary (SRTM-class, ~30 m cells) crossing the park, not real local topography. The CAD ground truth was correctly deferred to — all sections are drawn flat at z=0.

5.3 Key/Location Plan

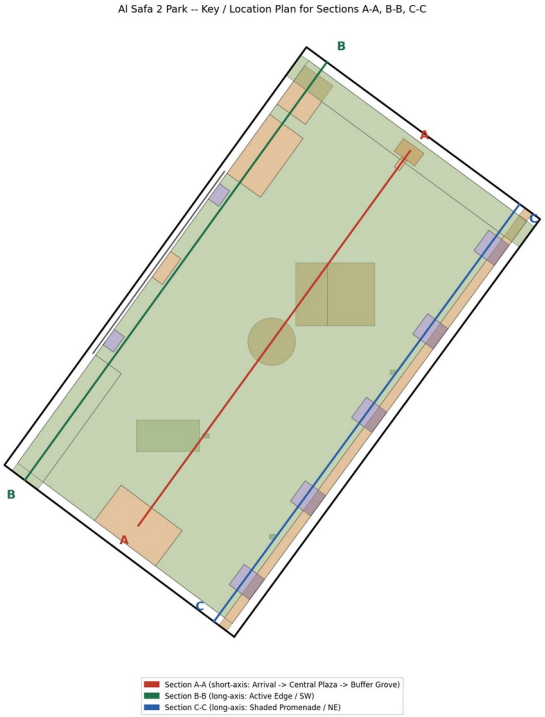


Figure 5.1 — Section-cut locations on the confirmed masterplan.

[VERIFIED] Section A-A's three anchor points (Arrival Plaza, Central Gathering Plaza, Buffer Grove) were confirmed colinear directly from their coordinates — 0.02 m deviation at the midpoint, not eyeballed. Sections B-B and C-C are developed (unrolled) running sections along their respective edges, explicitly flagged as such since elements sit at varying depth from the edge rather than on one rigid perpendicular plane — standard landscape-architecture practice for a linear zone, not a shortcut.

5.4 Section A-A — Arrival Plaza → Central Gathering Plaza → Buffer Grove

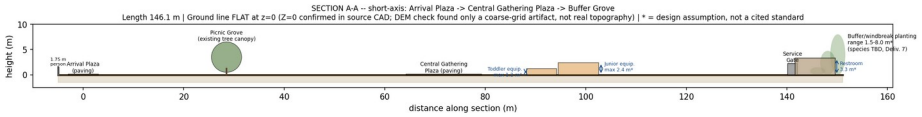


Figure 5.2 — Section A-A, 146.1 m. Within an 8 m tolerance band, this line also incidentally crosses Picnic Grove, Nature Play 3, the Event Lawn, both Playground zones, the Service/Utility Gate, and the Restroom — all shown, since a real cutting plane shows what it actually intersects rather than only the 3 named anchors.

5.5 Section B-B — Active Edge (SW), Mosque Buffer to Courts

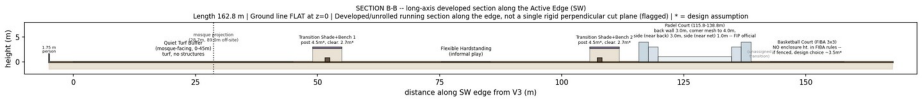


Figure 5.3 — Section B-B, 162.8 m, developed running section. Shows the mosque projection point (28.7 m from V3), the Quiet Turf Buffer, both transition structures, the Flexible Hardstanding, and the Padel/Basketball courts with corrected FIP wall-height zoning (see 5.7).

5.6 Section C-C — Shaded Promenade (NE)

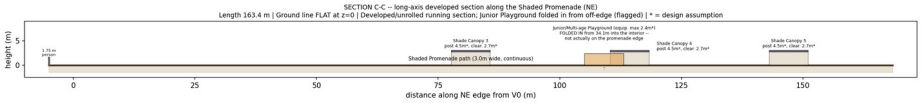


Figure 5.4 — Section C-C, 163.4 m, developed running section through 3 of the 5 shade canopies.

[NOTE] The Junior/Multi-age Playground is not actually on the promenade edge — its centroid sits 34.1 m into the interior. It is folded into the running section at its along-edge projection point, clearly labeled as a fold rather than silently placed on the edge line where it doesn't belong.

5.7 Heights — Sourced, One Correction Made

[CORRECTED] Independently cross-checked the Padel court wall-height claims against multiple current sources (padel governing-body specifications reported consistently across several independent references). The back-wall figure (4.0 m total: 3.0 m glass + 1.0 m mesh) is correct. However, the side-wall claim of "~1.0 m near the net" does not match any source found — every independent reference describes side walls as 4.0 m for the first ~2 m from the back corners, stepping down to 3.0 m for the remaining length toward the net, not down to 1.0 m. The table below uses the corrected 3.0 m figure. This looks like a genuine sourcing error in the original pass rather than a legitimate FIP variant — worth Claude Code double-checking its source next time a sports-facility standard is cited.

Element	Height	Source	Design assumption?
Padel — back wall	3.0 m glass + 1.0 m mesh = 4.0 m total	FIP (International Padel Federation)	No
Padel — side wall, near back corners	4.0 m	FIP	No
Padel — side wall, toward net	3.0 m (corrected — see above)	FIP, cross-verified against multiple sources	No
Basketball (FIBA 3x3) — enclosure	No standard — open court	FIBA official 3x3 rules — independently confirmed, no fencing/enclosure clause exists	If fenced, ~3.5 m would be a design choice only

Element	Height	Source	Design assumption?
Shade Canopy — post height / clearance	4.5 m post / 2.7 m clearance	Typical commercial shade-sail practice	Yes — not a formal code
Toddler Playground equipment — max height	1.2 m	EN 1176 (equipment) / EN 1177 (safety surfacing) framework — the standard recognized by Dubai Municipality, Trakhees, and KHDA; figure is common industry practice within it, not a mandated cap	Yes
Junior/Multi-age Playground equipment — max height	2.4 m	Same EN 1176/1177 framework, 5–12 yr band	Yes
Restroom / Comfort Station — building height	3.3 m	Typical small public-park restroom practice	Yes — no specific code cited
Bench — seat height	0.45 m	Widely used public-seating convention	No
Bench — overall height (with backrest)	0.82 m	Typical park-bench product dimension	Yes
Windbreak / buffer planting	1.5–8.0 m (shrub to tree)	General arid-climate windbreak knowledge; species TBD	Yes — placeholder pending Deliverable 7
Perimeter boundary wall/fence	Undetermined	Confirmed absent from the Scope of Work (checked directly)	N/A — flagged as absent

5.8 Open Items

- **Buffer planting height** is a placeholder range pending species selection in Deliverable 7 — not a design commitment.
- **Boundary wall/fence height** remains genuinely undetermined — a structural decision for later, not an oversight here.
- **Playground and shade-structure heights** are industry practice, not mandated ceilings — flagged consistently in-table rather than presented as fixed code minimums.

Sources: Deliverables 1–4 (this project); International Padel Federation (FIP) court specifications; FIBA 3x3 Official Rules; EN 1176 / EN 1177 (playground equipment and safety-surfacing standards recognized by Dubai Municipality, Trakhees, and KHDA — corrected from an earlier CPSC citation, see Deliverable 7). Section geometry, ground-truth elevation check, and initial sourcing: Claude Code. Independent height-standard verification and correction: this document. Full AI-tool-by-tool methodology log: Deliverable 10 — AI Methodology Report.